Types

1. **Complete fracture** involves the break across the entire cross section of the bone and is frequently displaced.
2. **Incomplete Fracture:** The break occurs only through part of the cross section of the bone.
3. **Open fracture:** The skin may be pierced by the bone or by a blow that breaks the skin at the time of the fracture. The bone may or may not be visible in the wound.
4. **Closed Fracture:** Does not produce a break in the skin.
5. **Pathologic fracture**: A pathologic fracture (also called insufficiency fracture) is a bone fracture caused by disease that led to weakness of the bone structure.

Pattern of fracture

- **Transverse fracture:** This type of fracture has a horizontal fracture line.
- **Oblique fracture:** This type of fracture has an angled pattern.
- **Comminuted fracture:** In this type of fracture, the bone shatters into three or more pieces.
- **Other types:** Avulsion fracture, Compression (crush) fracture, Greenstick fracture, Hairline fracture, Impacted fracture, Intra articular fracture, Longitudinal fracture, Spiral fracture, Stress fracture, Torus (buckle) fracture.

Causes

Most fractures are caused by

- Fall
- Automobile accident.
- Osteoporosis, infection
- Tumor
- Direct violence
- Bending forces
- Crushing force
- Sudden twisting motion.

Signs and symptoms

- Pain, swelling, bruising, discoloration of the skin
- Angulation
- False motion
- deformity
- Shortening and Crepitus
- Tenderness
- The patient is unable to put weight on the injured area
- The patient cannot move the affected area
- Pale and clammy skin
- Dizziness

Diagnosis

1. History of incident whether fall, accident, trauma should be asked to patient or witnesses.
2. Physical examination, identify signs and symptoms, and make a diagnosis.
3. X-ray, CT scans, MRI done as required.

The treatment of fractures

Goals of fracture treatment

- Restore fracture fragments to their normal anatomic position (Reduction)
- Maintain reduction in place until heating occurs (Immobilization)
- Promote regaining of normal function and strength of the affected part (rehabilitation)

Methods for obtaining fracture reduction

- Closed reduction
- Traction
- Open reduction

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